Latent Collapse Diagnostics and MHA Ablations

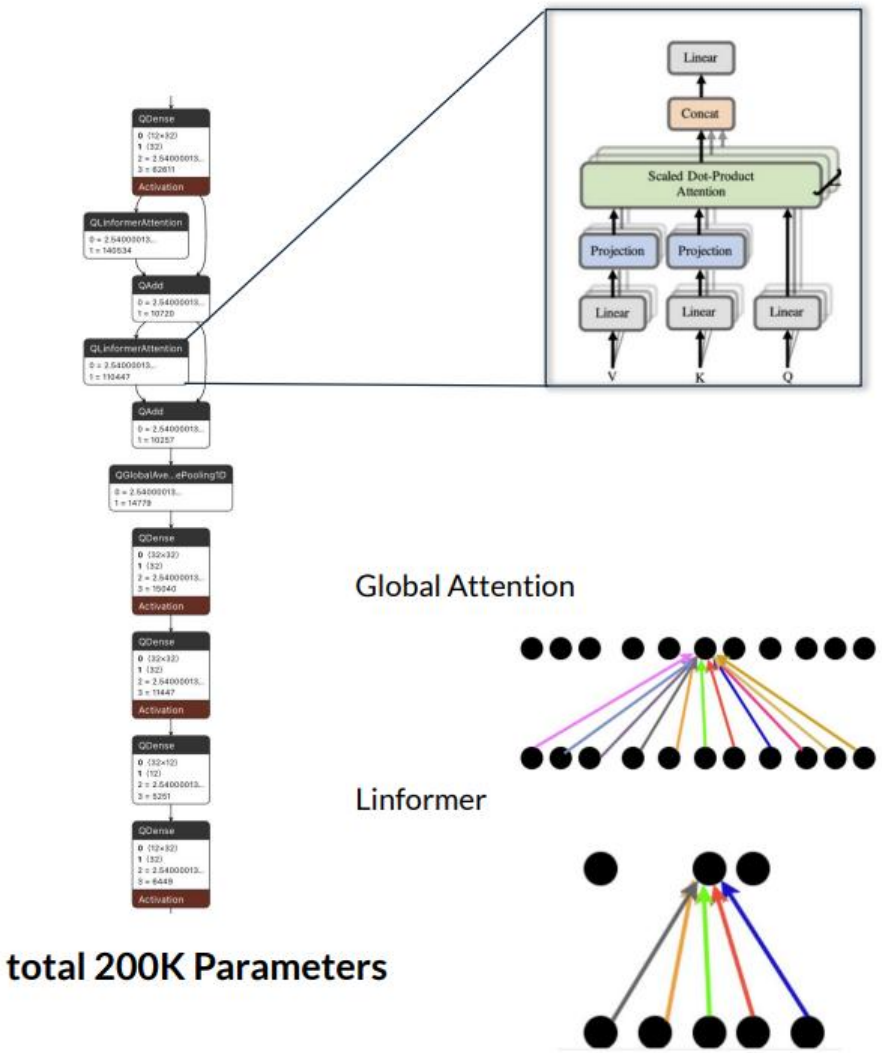
L1-only particle-level embedding model for anomaly detection

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the NextGen WP3 OpenLab -

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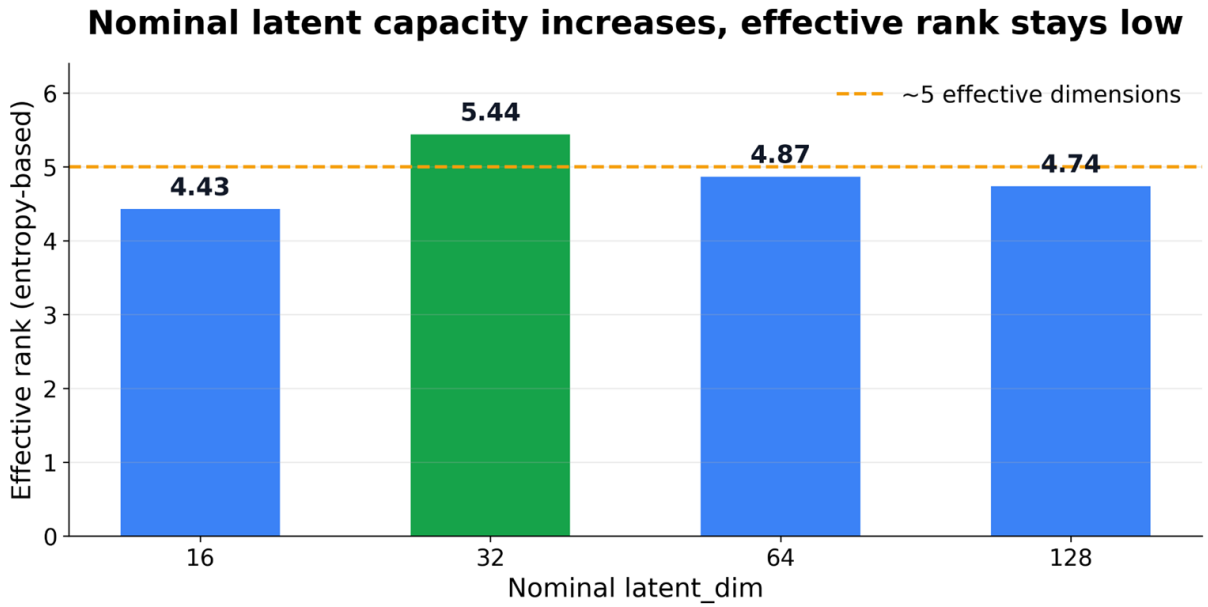
Project Reminder

- Input — Updated Collide-2V L1-only candidates (no offline reconstruction).
- Per-candidate features — pt, eta, phi, D0/dxy, PID.
- Candidate selection — top-pT L1 candidates per event (up to 100 constituents).
- Task — learn one event-level latent embedding, then classify it.
- Classes — MinBias, Top, QCD, DY, HH4B.
- HH4B is held out conceptually as the anomaly benchmark against the full SM background (MinBias+Top+QCD+DY).



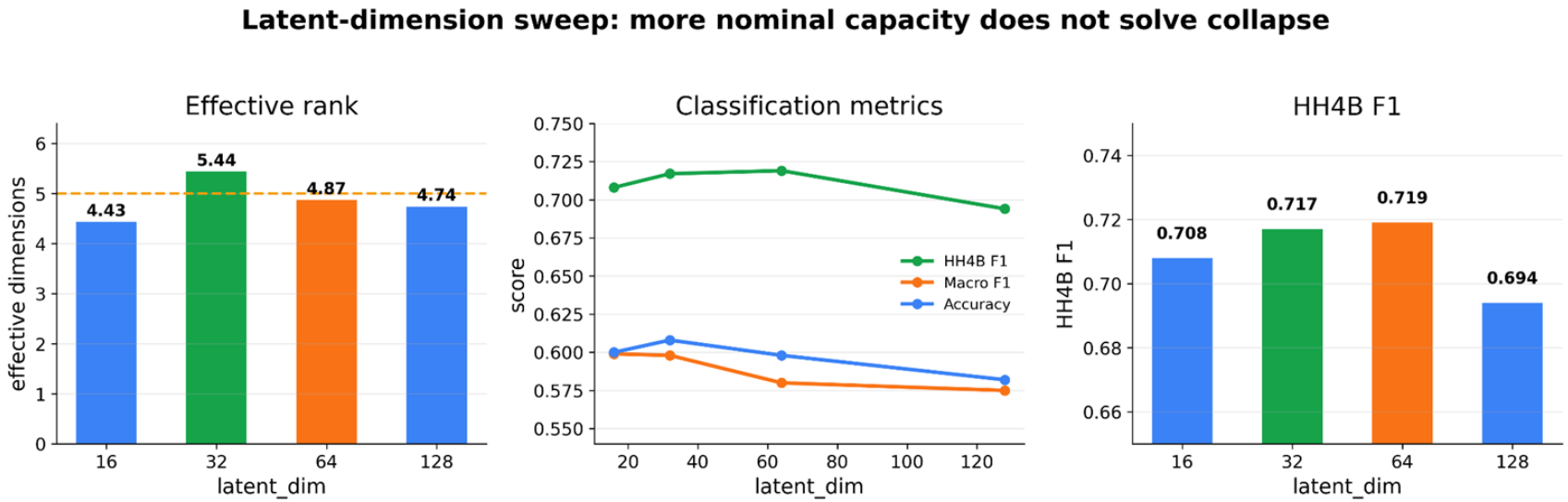
Latent Collapsing?

- Increasing latent_dim from 16 → 32 → 64 → 128 did not consistently improve accuracy or HH4B F1.
- **Effective dimensionality stayed low** across the entire sweep — roughly 4–5 active directions, regardless of nominal size.
- PCA / covariance diagnostics show most latent variance concentrated in only a few directions.
- **This suggests nominal latent capacity is not the real bottleneck.**



The model is given 8x more latent capacity, but it still uses only about 4-5 effective dimensions.

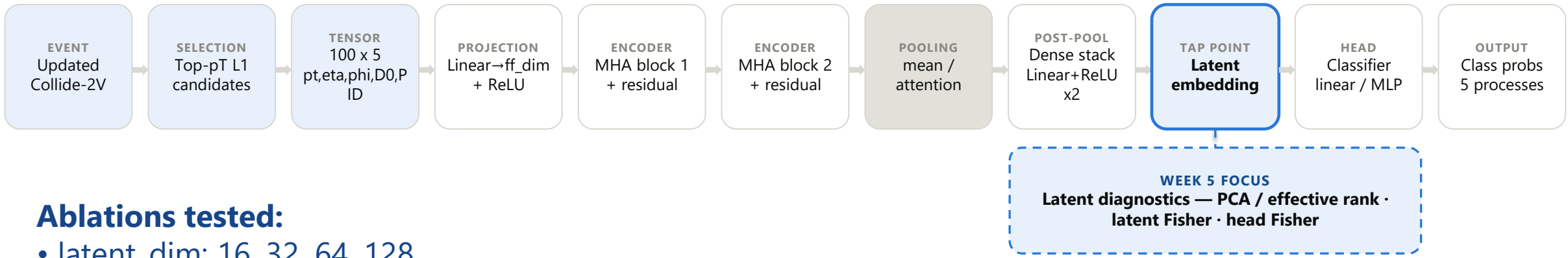
latent_dim = 32 is the best-balanced configuration



LATENT_DIM	PARAMS	VAL ACC	MACRO F1	HH4B F1	DY F1	EFF. RANK	PARTIC. RATIO	PCS@90%	EVAL_Q
16	181,526	0.602	0.599	0.708	0.456	4.43	3.09	4	0.169
32 ★ best balance	**185,638**	**0.609**	**0.598**	**0.717**	**0.400**	**5.44**	**3.36**	**5**	**0.193**
64 ⚠ unstable eval_q	193,862	0.598	0.580	0.719	0.315	4.87	2.88	5	0.00004
128	210,310	0.585	0.575	0.694	0.365	4.74	2.98	5	0.176

MHA event encoder — where the diagnostics attach

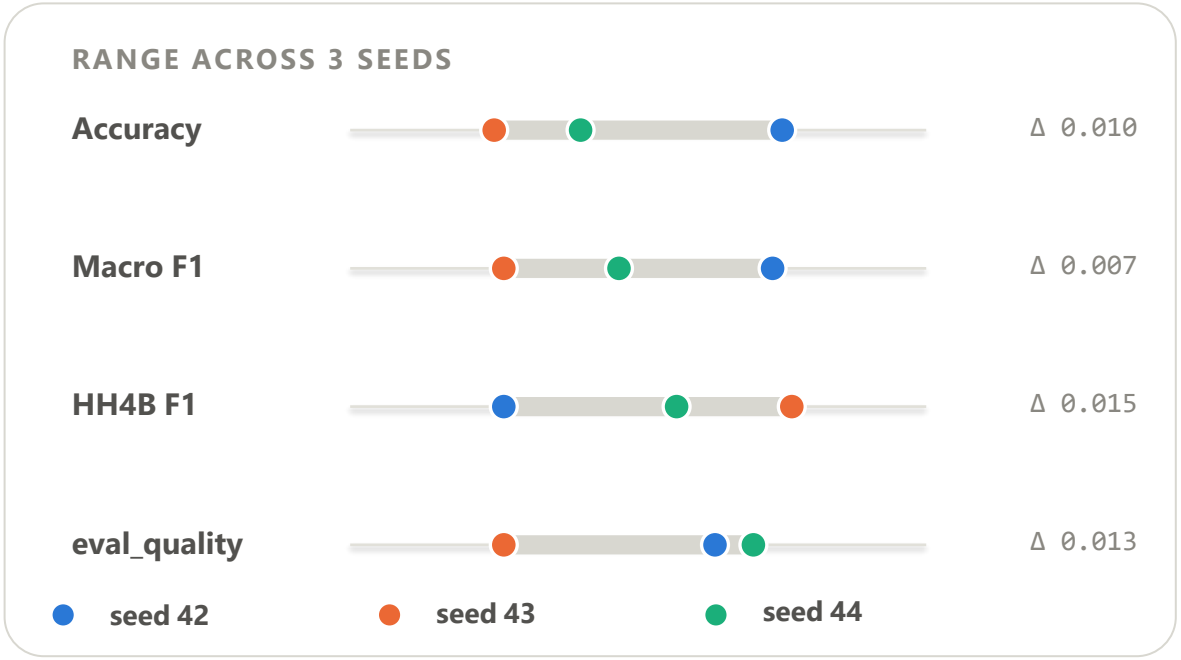
- L1-only input: top-pT PUPPI candidates, 100 × 5 features.
- MHA encoder with residual attention blocks.
- Event-level latent embedding used for classification and diagnostics.



Ablations tested:

- latent_dim: 16, 32, 64, 128
- pooling: mean vs attention
- classifier head: linear vs MLP
- attention blocks
- ff_dim
- num_heads
- CrossEntropy + latent variance regularization

Repeated-seed stability at latent_dim = 32



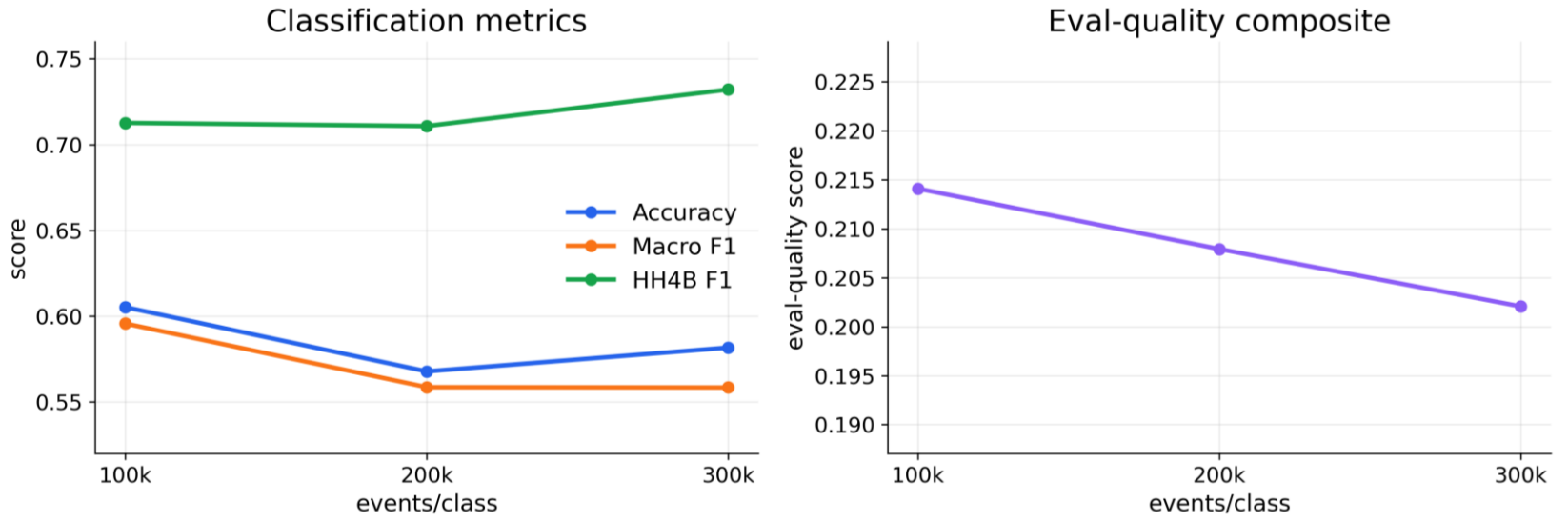
SEED	ACC.	MACRO F1	HH4B F1	EFF. RANK	PARTIC. RATIO	EVAL_Q
42	0.605	0.596	0.713	5.04	3.22	0.214
43	0.595	0.589	0.728	4.69	2.79	0.203
44	0.598	0.592	0.722	4.66	2.83	0.216

- Repeated the 100k/class, latent_dim=32 configuration with seeds 42, 43, and 44.
- **Accuracy, macro F1, HH4B F1, and eval_quality were stable** across all three seeds.

All three seeds cluster tightly across every metric — the observed behavior is a property of the setup, not one lucky (or unlucky) initialization.

More data does not remove the collapse

EVENTS/CLASS	TOTAL EVENTS	ACCURACY	MACRO F1	DY F1	HH4B F1	EFF. RANK	PARTIC. RATIO	EVAL_Q
100k	500k	0.605	0.596	0.396	0.713	5.04	3.22	0.214
200k	1.00M	0.568	0.559	0.494	0.711	3.37	2.43	0.208
300k	1.50M	0.582	0.558	0.230	0.732	2.88	1.90	0.202



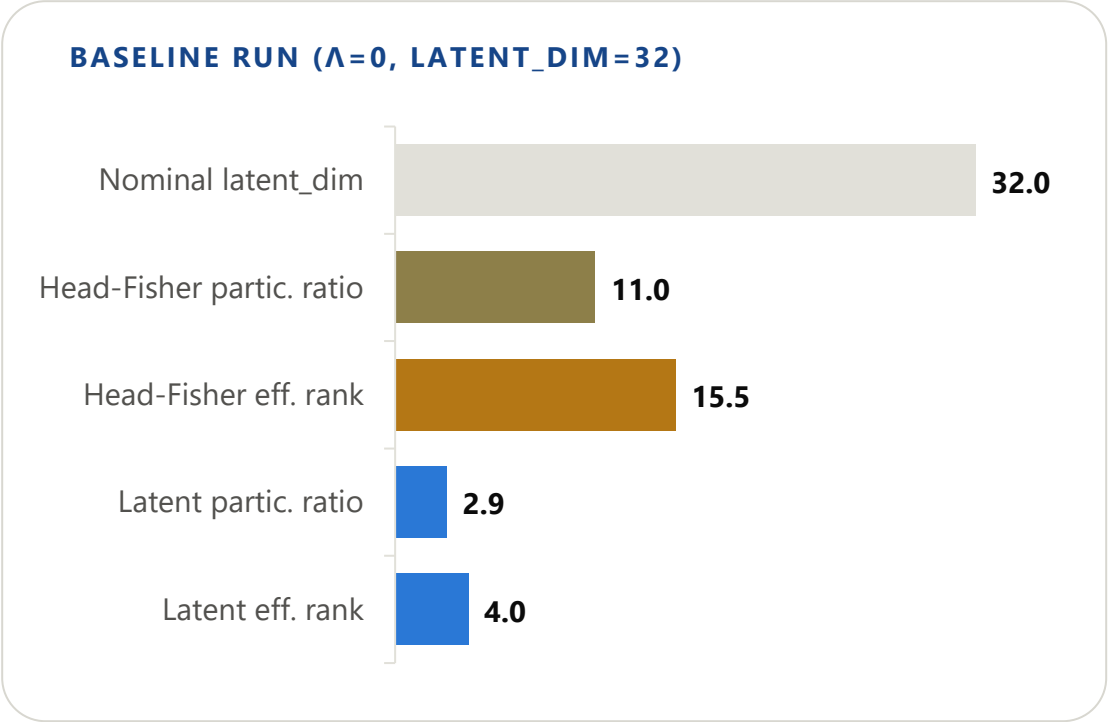
More data alone does not produce a monotonic improvement in HH4B-vs-SM performance or eval-quality.

No single architecture knob removes the collapse

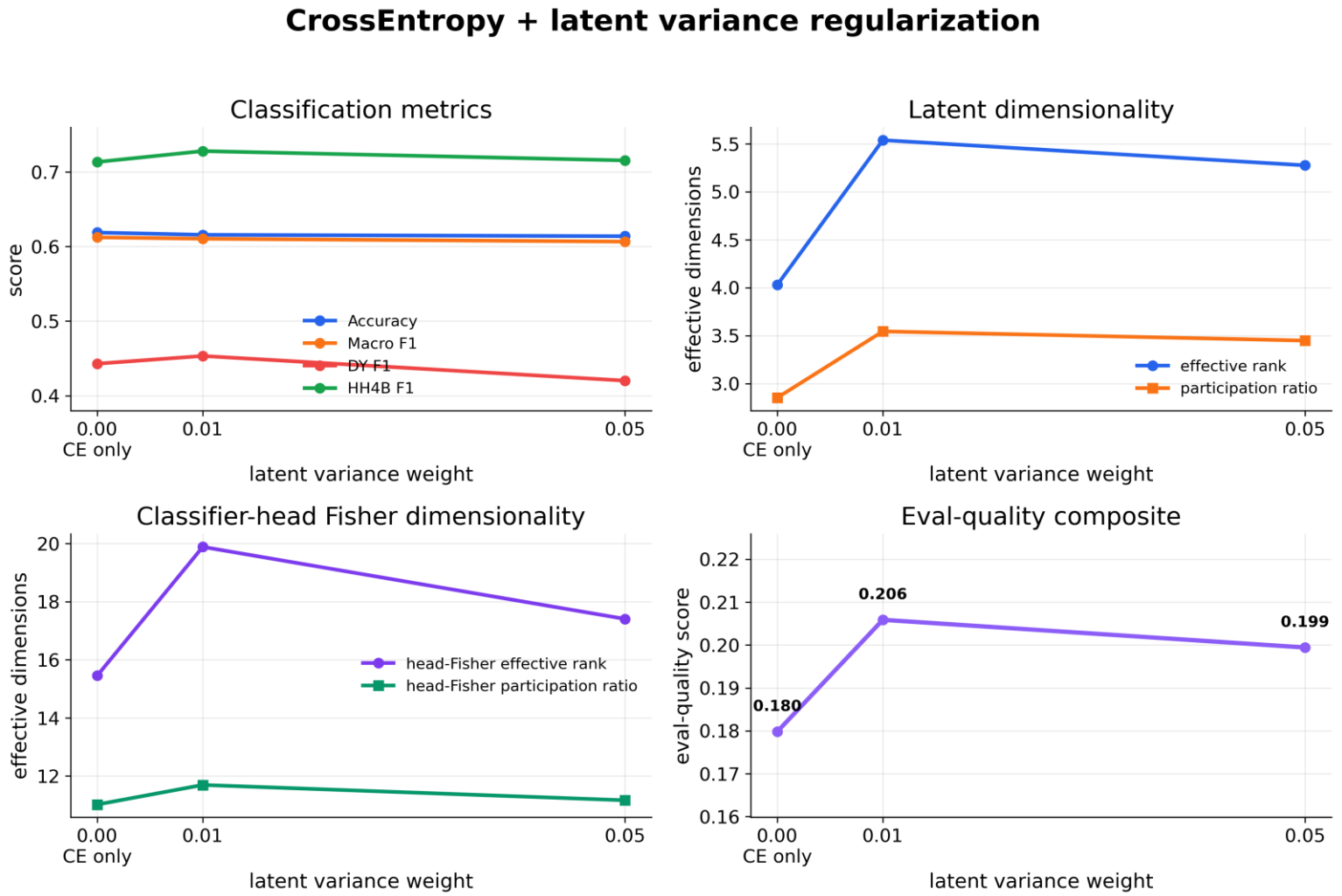
ABLATION AXIS	SETTING	ACCURACY	MACRO F1	HH4B F1	EFF. RANK	EVAL_Q
Pooling	attention	0.605	0.596	0.713	5.04	0.214
Pooling	mean	0.612	0.610	0.722	6.26	0.208
Attention blocks	1	0.609	0.609	0.723	3.65	0.213
Attention blocks	2 (baseline)	0.619	0.612	0.713	4.03	0.180
Attention blocks	4	0.554	0.553	0.705	1.48	0.229
ff_dim	64	0.603	0.599	0.717	2.95	0.213
ff_dim	128 (baseline)	0.619	0.612	0.713	4.03	0.180
ff_dim	256	0.620	0.610	0.733	4.60	0.123
num_heads	4	0.610	0.601	0.714	3.89	0.197
num_heads	8 (baseline)	0.619	0.612	0.713	4.03	0.180
num_heads	16	0.616	0.609	0.709	4.01	0.191

The classifier uses more directions than PCA suggests

- **PCA / eigenvalue diagnostics** measure how much the latent representation varies geometrically.
- **Latent Fisher** estimates which latent directions actually affect the model's output.
- **Head Fisher** checks which classifier-head directions are functionally important.
- The classifier uses more directions than PCA alone suggests — but still far fewer than the nominal latent_dim.



CrossEntropy + latent variance regularization



A small variance penalty (lambda=0.01) increases latent/head-Fisher dimensionality and improves HH4B F1, while a stronger penalty starts to trade off DY performance.

Where this goes next

Increasing latent_dim alone is not enough — the representation stays low-dimensional under architecture and data-size changes alike.

The most promising direction going forward is to modify the learning objective — not only increase model capacity.

- latent_dim sweep: flat
- data-size sweep: flat
- architecture ablations: flat
- λ=0.01 variance term: promising